

FİZ1951-MUHENDİSLER İÇİN YARIİLETKEN FİZİĞİ ARA-2 FORMÜL KAĞIDI

$$\varepsilon_o = 8.85 \times 10^{-12} \frac{F}{m}$$

$$\mu_o = 4\pi \times 10^{-7} T.A/m = 1.257 \times 10^{-6} H/m$$

$$q = e = 1.6 \times 10^{-19} C$$

$$h = 6.63 \times 10^{-34} J.s \quad m_o = 9.11 \times 10^{-31} kg$$

$$k_B = 1.38 \times 10^{-23} J/K \quad 1eV = 1.6 \times 10^{-19} J$$

$$c = 3 \times 10^8 \frac{m}{s}$$

$$H = \frac{NI}{l}$$

$$M = \chi_M H$$

$$B = \mu_o H + \mu_o M$$

$$H = \frac{B}{\mu}$$

$$\mu = \mu_o \mu_r = \mu_o (1 + \chi_M)$$

$$R = \frac{(n_r - 1)^2 + k^2}{(n_r + 1)^2 + k^2}$$

$$\vartheta = \frac{1}{\sqrt{\mu\varepsilon}} = \frac{c}{n} \quad n_r = \frac{c}{\vartheta}$$

$$\alpha(h\nu) = A(h\nu - E_g)^{\gamma}$$

$$I(d) = I_o \exp(-\alpha d)$$

$$T = (1 - R^2) \exp(-\alpha d)$$

$$I_D = I_{DSS} \left(1 - \frac{V_{GS}}{V_{GS(off)}}\right)^2$$

$$I_E = I_C + I_B$$

$$I_E = \frac{\beta + 1}{\beta} I_C$$

$$I_C = \alpha I_E$$

$$\alpha = \frac{\beta}{\beta + 1}$$

$$I_E = (I_S / \alpha) e^{V_{BB}/V_T}$$

$$I = I_o \left(\exp \frac{(qV_{D_{IS}})}{nk_B T} - 1 \right) \cong I_o \left(\exp \frac{(qV_{D_{IS}})}{nk_B T} \right)$$

$$np = n_i^2$$

$$V_{bi} = \frac{1}{2} E_{max} W$$

$$V_{bi} = \frac{kT}{e} \ln \frac{p_p n_n}{n_i^2}$$

$$E = \frac{hc}{\lambda} \quad E = \frac{1.24 eV \cdot \mu m}{\lambda(\mu m)}$$

$$E = \frac{1240 eV \cdot nm}{\lambda(nm)} \frac{1}{\mu_n^T} = \frac{1}{\mu_n^L} + \frac{1}{\mu_n^I}$$

$$\sigma_p = pq\mu_p$$

$$\sigma_n = nq\mu_n$$

$$\sigma = \frac{ne^2 \tau}{m_e^*}$$

$$\vartheta_t = \sqrt{\frac{3k_B T}{m^*}}$$

$$W_o = \sqrt{\frac{2\varepsilon_o \varepsilon(V_{bi})}{en}} \quad (V_{d_{IS}} = 0)$$

$$W = \sqrt{\frac{2\varepsilon_o \varepsilon(V_{bi} - V_{d_{IS}})}{en}} \quad (V_{d_{IS}} > 0)$$

$$W = \sqrt{\frac{2\varepsilon_o \varepsilon(V_{bi} - (-V_{d_{IS}}))}{en}} \quad (V_{d_{IS}} < 0)$$

$$eV_{bi} = E_g + kT \ln \frac{N_A N_D}{N_C N_V}$$

$$W_p^2 = \frac{2\varepsilon}{e} V_{bi} \left[\frac{N_D}{N_A(N_D + N_A)} \right]$$

$$W_n^2 = \frac{2\varepsilon}{e} V_{bi} \left[\frac{N_A}{N_D(N_D + N_A)} \right]$$

$$W = \left(\frac{2\varepsilon}{e} V_{bi} \left[\frac{N_A + N_D}{(N_D N_A)} \right] \right)^{1/2}$$

$$E_{max}(x=0) = -\frac{e}{\varepsilon} N_D W_n = -\frac{e}{\varepsilon} N_A W_p$$

$$\vec{x} = \vec{v}_s \Delta t$$

$$I = \frac{\Delta Q}{\Delta t}$$

$$I = nqA\vartheta_s$$

$$J = \frac{I}{A}$$

$$\vec{J} = nq\vec{\vartheta}_s$$

$$V = IR$$

$$R = \rho \frac{l}{A}$$

$$\vec{v}_s = \mu \vec{E} = \frac{q\tau}{m} \vec{E}$$

$$\vec{J} = nq\mu \vec{E}$$

$$\sigma = nq\mu = ne\mu$$

$$q = e$$

$$\mu_n = \frac{e\tau_n}{m_n^*}$$

$$\mu_p = \frac{e\tau_p}{m_p^*}$$

$$\vec{J} = \sigma \vec{E} \quad \vec{E} = \frac{V}{l}$$

$$\rho = \frac{1}{ne\mu}$$

$$l = \mu_{ort} \tau$$

$$\mu_{th} = l / \tau$$

$$\sigma = 1/\rho \rightarrow S = 1/R$$

$$\vec{F}_e = q\vec{E} = m_o \vec{a}$$

$$\vec{F}_e = q\vec{E} + q\vec{E}_{Kr} = m^* \vec{a}$$

$$V_{CB} = V_{CE} - V_{BE}$$

$$V_{CE} = V_{CC} - V_{RC}, \quad V_{RC} = I_C R_C$$

$$V_{RB} = V_{BB} - V_{BE}, \quad V_{RB} = I_B R_B$$

$$I_C = I_S (e^{V_{BB}/V_T} - 1), I_C \gg I_S \text{ şartı ile}$$

$$I_C = I_S e^{V_{BB}/V_T} \text{ dir.}$$

$$I_S = \frac{A_E q D_n n_{p0}}{W}, \quad n_{p0} = n_i^2 / N_A$$

$$V_T = \frac{kT}{q}$$

$$\beta = \frac{I_C}{I_B}$$

$$I_B = \frac{I_C}{\beta}$$

$$\beta = \frac{\alpha}{1 - \alpha}$$

$$V_T \cong 26 \text{ mV (@300K)}$$

$$I_B = \left(\frac{I_S}{\beta} \right) e^{V_{BB}/V_T}$$

$$I_E = \frac{\beta + 1}{\beta} I_S e^{V_{BB}/V_T}$$